Section VI — Ground Support Equipment

Replaces	HCR-5100 rev 01, pages 235–283 (§6.1, §6.2, §6.3 OP 6 and OP 8)
Revision	rev 02 draft — ECR-01, control subsystem change
Date	6 November 2026
Reason	Replace the FS-i6X / FS-iA10B radio pair with a laptop-driven ESP32-S3 + DX-LR41-24T12S link
Status	Draft for review — not to be used for a live firing until the test gates in the companion test plan are signed

How to use these pages

This is an insert, not a rewrite of the whole manual. Pages numbered **236–255** replace the existing Section VI pages; the content that follows them (Section VII) needs the four changes listed at the end of this insert, which are also summarised in the revision history entry below. Everything not mentioned here — the fluid system, the servo-actuated ball valves, the cables, the fill line, and all of Section V — is unchanged.

Revised content is marked ■ **REVISED**. Deleted text is shown struck through only where a reviewer needs to see exactly what was removed.

Revision history addition

Revision	Description	Date
01	Initial release	17 Aug 2024
02	GSE control subsystem replaced (ECR-01): FS-i6X transmitter and FS-iA10B receiver replaced by a laptop-driven ESP32-S3 pair with DX-LR41-24T12S 2.4 GHz LoRa modules. Section VI §6.1, §6.2, §6.3 OP 6 and OP 8 revised; §7.2, §7.3 and §7.5 amended.	6 Nov 2026

What changes, at a glance

Item	Rev 01	Rev 02
Operator control	FS-i6X transmitter, four switches under covers	Laptop application over USB-C, plus a physical arm key at the pad
Radio link	AFHDS 2A, 2.4 GHz GFSK, closed protocol	2.4 GHz LoRa (SX1281), 13 dBm, full-duplex by channel split
Receiver	FS-iA10B, 10 PWM channels	ESP32-S3 pad node, 3 PWM channels + igniter relay
Channel map	Ch5 / Ch6 / Ch8 / Ch3, fixed by the transmitter firmware	Same numbers, software-defined, re-mappable from the GUI
Endpoint adjustment	Transmitter menu, ±120 %	GUI, servo travel 5–90° per channel

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Failsafes	Per-channel positions in the transmitter menu	Pad-node watchdog (0.3–3.0 s) + TX-node safe stream + hardware heartbeat
Igniter	PWM relay, on-time set by the operator's finger	One pulse per press, hard limit 0.1–2.0 s, series key switch and heartbeat
Logging	None	Every command and every reported state, CSV, 20 Hz
Cost	—	+$16 net; −$72 of removed parts, +$89 of added parts

What the pad node does not do. It does not replace the fill valve, the QD, the fill line, the supply bottle, the servo-actuated ball valves, or their adjustment procedure. It is a drop-in replacement for the receiver and the four things the receiver used to switch.

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■ REVISED 6.1 GSE Overview

Ground Support Equipment, or **GSE**, is a critical part of any rocket system. The launch vehicle may be the most interesting part, but it simply does not function without GSE.

For solid rockets, GSE can be as simple as a switch that fires an igniter. For UC-valve hybrids, where the igniter also acts as the main oxidizer valve, it must also be capable of loading oxidizer into the motor. For Mojave Sphinx and all standard liquid rockets, it must also be able to open the main propellant valves to start the flow of fuel and oxidizer, separate from the ignition command. Furthermore, since any operation in which N_2O is present outside the supply bottle is hazardous to personnel, all such activities must be performed at a safe distance. This requires a remotely-controlled fluid handling system which is both safe and economical, so as to not dwarf the cost of the rocket itself.

The purpose of the GSE is to:

1. Control flow of N_2O from a supply bottle
2. Provide ignition power
3. Issue a valve-open command
4. Dump N_2O rapidly to safe the vehicle

For rockets with pneumatic Half Cat valves, the valve-open command may be a trigger for a pyrotechnic device, such as a line cutter that severs and vents a pneumatic pressure supply tube. In any standard liquid rocket, the oxidizer dump may be the same command as the normal valve-open, only without triggering the igniter, and may involve dumping both propellants. Only in the unlikely event of complete power loss during fill should the static vent be relied upon to fully empty the oxidizer tank, as this can take upwards of an hour depending on tank and vent sizes. The remote control system described in this guide is configured so that **a loss of communication with the GSE will automatically open the vehicle propellant valves**, returning the system to a safe state.

The GSE can be divided into two major subsystems: the fill system and the control system. The fill system is a simple plumbing assembly consisting of a supply bottle connection, a fill valve, and a QD connection.

The control system is purpose-built and laptop-driven. A small application on the operator's laptop sends command frames over USB-C to an ESP32-S3 radio node inside the GSE box, which relays them over a 2.4 GHz LoRa link to a second ESP32-S3 at the launch pad. The pad node drives the three valve servos and the igniter relay directly, and reports the actual state of every output back to the laptop ten times per second. Channel numbering is preserved from earlier revisions — FILL on Ch5, VEHICLE on Ch6, PURGE on Ch8, IGNITER on Ch3 — but is software-defined and can be re-mapped without re-wiring.

Because the link is not a certified hobby receiver, **all failsafe behaviour is implemented in the pad node** and verified as part of launch preparation: loss of communication for longer than the configured watchdog (1.0 s by default) closes the fill and purge valves, opens the vehicle valves, and de-energises the igniter. This mirrors the safe state that the FS-iA10B receiver was configured for in earlier revisions, and the physics of the safe state is unchanged: no pressure remains in the vehicle, and the igniter cannot fire.

The control system consists of the operator's laptop, two ESP32-S3 radio nodes, two DX-LR41-24T12S modules, the 12 V battery, the voltage regulator, the arm key switch, and the igniter relay. Below is the basic electrical schematic of the GSE.

Figure 6.1 (revised): GSE electrical schematic

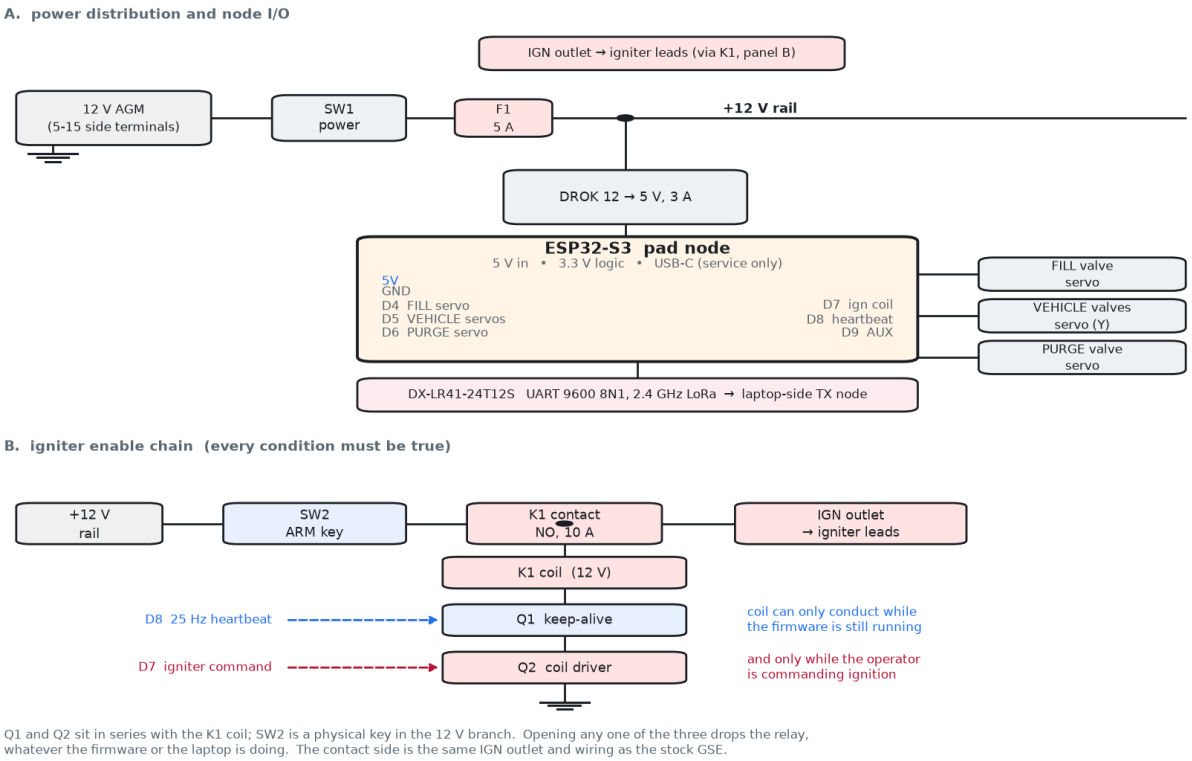


Figure 6.1: GSE electrical schematic (revised). Panel A — power distribution and node I/O. Panel B — igniter enable chain; the relay coil can only conduct with the arm key closed, the heartbeat running and an igniter command present.

Why the link is not a hobby receiver any more

The FS-iA10B listens on FlySky AFHDS 2A, a closed frequency-hopping protocol that a general-purpose 2.4 GHz LoRa module cannot produce. Removing the receiver therefore means taking ownership of the behaviour it used to provide for free: PWM generation, channel mapping, and failsafes. The pad node and its test plan exist to make that ownership explicit rather than implicit.

■ REVISED 6.2 GSE Bill of Materials

Line items 1–20 (fluid system and enclosure) are unchanged from rev 01 except where noted. Lines marked **NEW** are additions; lines marked **DELETED** are removed.

Line	Part Number	Description	Unit	Qty	Cost	Context
1	~~DTUS_FSI6X_10CH~~	~~FlySky FS-i6X 10CH transmitter w/ iA10B receiver~~	~~$59.99~~	~~1~~	~~$59.99~~	~~Controller~~
2	AZ18080602-03	Toggle switch guards, 12 mm (pack of 6) — retained for SW1/outlet labelling	$8.49	1	$8.49	Switch covers
3	62586	Thunderbolt Magnum 12 V 160 CCA AGM battery	$49.99	1	$49.99	Electrical connections
4	20GA-RBP	20 GA silicone wire, red/black, 10 ft each	$0.00	1	$0.00	Leftover from avbay
5	~~LCATMC1569~~	~~PWM-controlled relay module, 10 A 30 V~~	~~$11.99~~	~~1~~	~~$11.99~~	~~Ignition switch~~
6	090600AFA	DROK 12 V to 5 V converter, 3 A — retained (now also feeds the pad node)	$12.99	1	$12.99	Battery to logic
7	GSW-17	Gardner Bender SPST toggle switch, 6 A	$5.99	1	$5.99	GSE power switch
8	7159K3	Straight-blade receptacle, 5-15 NEMA, side terminal	$3.36	1	$3.36	IGN and service outlets
9	7526K23	Nylon wall plate for NEMA outlet	$0.73	1	$0.73	Outlet cover
10	YXQ-3PIN-1M	3-pin servo extension, 1 m (pack of 10) — 2 retained for the vehicle Y-harness	$0.00	1	$0.00	Leftovers from valves
11–37	Unchanged from rev 01 (toolbox, eye bolts, servo, ball valve, tank, fittings, tubing, filament, printed parts)				$333.98	

Added in rev 02

Line	Part Number	Description	Unit	Qty	Cost	Context
38	ESP32-S3-DEVKITC-1	ESP32-S3 devkit with native USB (or N16R8 / 8 MB equivalent)	$8.00	2	$16.00	TX node, pad node
39	DX-LR41-24T12S	2.4 GHz SX1281 LoRa module, UART, 13 dBm	$9.50	2	$19.00	Radio pair
40	ANT-24G-SMA-3DBI	2.4 GHz antenna, SMA, 3 dBi (with IPEX pigtail)	$3.00	2	$6.00	Antennas
41	PRINTED-CARRIER	Carrier board / proto board for the pad node, with servo headers and a 12 V screw terminal	$3.00	1	$3.00	Mounting
42	RLY-12V-1CH-OPTO	Relay module, 1-channel, 12 V coil, 10 A, opto-isolated, with flyback	$6.00	1	$6.00	Igniter switch (replaces line 5)
43	IRLZ44N		$1.50	3	$4.50	Interlocks

		Logic-level N-MOSFET (Q1 keep-alive, Q2 coil driver, one spare)				
44	1N5819	Schottky diode, coil flyback and RC pump	$0.20	4	$0.80	Transient protection
45	SW-KEY-SPST	Key switch, 2-position, SPST (SW2 arm key)	$6.00	1	$6.00	Physical arm interlock
46	FUSE-5A-INLINE	5 A blade fuse and inline holder (F1)	$3.00	1	$3.00	Over-current
47	CAP-100U-16V	Electrolytic 100 µF, plus 100 nF ceramic, module supply decoupling	$1.20	1	$1.20	RF stability
48	USB-C-PANEL-1M	Panel-mount USB-C extension, 1 m	$7.00	1	$7.00	Laptop connection
49	WIRE-KIT-22AWG	22 AWG hookup wire, crimp terminals, heat-shrink, servo crimp kit	$6.00	1	$6.00	Harness
50	PWRBANK-5V-2A	Optional USB power bank for the TX node	$12.00	1	$12.00	Redundancy

Subtotal	**Rev 01**	**Rev 02**	**Delta**
Removed lines 1 and 5	$71.98	$0.00	−$71.98
All other lines	$457.47	$457.47	$0.00
New lines 38–50	$0.00	$86.50	+$86.50
GSE total	**$529.45**	**$543.97**	**+$14.52**

The operator's laptop substitutes for the transmitter and is excluded, as the guide already assumes a laptop for data reduction (Section 3). Prices are indicative, single quantity, November 2026.

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6.3 GSE Assembly Procedures

The GSE presented in this section is constructed within a metal toolbox, which provides a durable housing that protects the system from exposure to the rocket's exhaust and can conveniently also store the tools needed to recycle the rocket between launches. The integrated packaging of the entire system including fill, control electronics, power, and optional CO_2 purge requires minimal on-site setup time and is well suited to long term use. If project budget is a limiting factor and only a small number of launches are planned, the GSE can be simplified at the expense of reduced ease of use by eliminating the toolbox and accompanying fittings and fasteners.

OP	Operation	Status in rev 02
OP 1	Assemble Servo-Actuated Ball Valves (GSE)	Unchanged
OP 2	Assemble Fill System	Unchanged
OP 3–5	Enclosure, bulkheads, fluid system integration	Unchanged
OP 6	Electrical System Integration	**REVISED** — see page 241
OP 7	Fabricate Cables and Fill Line	Unchanged
OP 8	Transmitter Setup & Firmware Configuration	**REPLACED** by OP 9 and OP 10
OP 9	Assemble and Wire the Control System	**NEW**
OP 10	Node Bring-up, Configuration and Checkout	**NEW**

OP 1: Assemble Servo-Actuated Ball Valves (GSE)

QTY	Part Number	Description
1	SABV-MOUNT-V2	Servo Valve Mount V2.STL
1	SABV-HCON-V2	Servo Valve Handle Connector V2.STL
—	N/A	3D printer filament (PLA+, PETG, or ABS recommended)
1	4112T22	Compact High Pressure Brass Ball Valve
1	DS3225	25 kg servo with 25T servo horn
1	93475A210	M3 flat washer, 7 mm OD
1	92855A310	M3 × 0.5, 10 mm low-profile socket head screw
1	90272A194	#8-32 pan head screw, ½"
1	90480A009	#8-32 hex nut
—	—	Nylon zip tie, 0.188" width
4	92180A044	M4 × 0.70, 16 mm hex head screw
4	90591A141	M4 × 0.7 hex nut
4	98689A113	M4 flat washer
4	92148A160	M4 split lock washer

The SABV-04FF servo-actuated ball valve assemblies used in the GSE are identical to those used as the fuel and oxidizer valves on the vehicle. Refer to Section 5 OP 1 for detailed instructions.

If constructing Fill Only GSE, only QTY 1 SABV-04FF is required. If a CO_2 purge valve for static testing is desired, build QTY 2 SABV-04FF.

Change note. The servo horn must be installed in the correct orientation, which in rev 01 was established with the transmitter's End Points menu (OP 8). In rev 02 the equivalent adjustment is made from the GSE application, and it must still be performed before the horn is finally screwed to the handle. See OP 10, step 5. Nothing else in OP 1 changes.

OP 2: Assemble Fill System

Unchanged from rev 01. Gather:

QTY	Part Number	Description
1	SABV-04FF	Servo-Actuated Ball Valve Assembly
1	79215A673	CGA 326 to 1/4 NPT adapter, brass
1	79215A672	CGA 326 female nut, brass
1	50785K92	High-pressure brass straight connector, 1/4 NPT female
2	50675K162	37° flared adapter, 1/4" tube OD × 1/4 NPT male

Follow rev 01 steps 1–8 without modification. OP 3, OP 4 and OP 5 are likewise unchanged.

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■ REVISED OP 6: Electrical System Integration

QTY	Part Number	Description
1	ESP32-S3-DEVKITC-1	ESP32-S3 devkit — **PAD** node
1	DX-LR41-24T12S	2.4 GHz LoRa module
1	ANT-24G-SMA-3DBI	2.4 GHz antenna
1	PRINTED-CARRIER	Pad-node carrier board
1	RLY-12V-1CH-OPTO	12 V, 10 A relay module
1	SW-KEY-SPST	Arm key switch (SW2)
3	IRLZ44N	Logic-level MOSFET
2	1N5819	Schottky diode
1	FUSE-5A-INLINE	5 A fuse and holder
1	CAP-100U-16V	100 µF + 100 nF, module decoupling
1	USB-C-PANEL-1M	Panel-mount USB-C extension
1	090600AFA	DROK 12 V to 5 V regulator (retained)
1	GSW-17	SPST power switch (retained)
1	7159K91	NEMA outlet (retained)
1	7526K53	Nylon wall plate (retained)
4	9027A194	#8-32 × ½" screws
—	WIRE-KIT-22AWG	22 AWG wire, crimp terminals, heat-shrink

1. Mount the pad-node carrier board on the inside of the back face of the toolbox, in the position formerly occupied by the FS-iA10B receiver and the PWM relay, using QTY 4 9027A194 screws with washers, lock washers and nuts. The carrier must be rigid: it carries the servo connectors and the 12 V ignition terminal. **Do not** mount the ESP32-S3 in a breadboard.
2. Fit the DX-LR41-24T12S to the carrier. Connect 3V3 to the module VCC, GND to GND, D16 to module TXD, D17 to module RXD, and D18 to module AUX. **Do not connect the module to 5 V.** Fit the 100 µF and 100 nF capacitors across the module supply pins.
3. Fit the antenna to the module, and route the SMA bulkhead through the existing antenna cover hole in the back face of the toolbox. The cover `CVR-ANT-2X8C` may be re-used by replacing the two wire antennas with a single SMA bulkhead; keep the antenna at least 30 mm from the servo harness.
4. Wire the three servo outputs to the carrier headers, observing the pin order used by the vehicle harness: signal → **D4** (FILL), **D5** (VEHICLE, both servos via the existing Y-harness), **D6** (PURGE); +5 V from the DROK regulator to the header power pins; ground common to the regulator ground.
5. Wire the ignition branch:
    1. 12 V battery positive → **F1** (5 A) → **SW1** → DROK regulator input and the ignition branch;

2. ignition branch → **SW2** arm key → relay module contact (NO) → the IGN NEMA outlet (longer slot), exactly as the rev 01 relay was wired;
3. relay module coil − → **Q2** drain, whose gate is **D7** and whose source is battery negative;
4. **Q1** in series between Q2 source and ground, gate driven from **D8** through the RC pump shown in Figure 6.1 panel B;
5. 1N5819 flyback across the relay coil terminals, cathode to +12 V.

6. Label the arm key "ARM — OPEN BEFORE APPROACHING PAD" and mount it on the toolbox face, adjacent to SW1. The key must be removable only in the OFF position.
7. Install the panel-mount USB-C extension through the toolbox face, adjacent to the power switch, and route it to the TX node. The TX node itself is carried with the laptop, not mounted in the box.
8. Retain the 12 V battery, the eye bolts, the DROK regulator and the outlet. The outlets are labelled IGN and 5V SERVICE; the former SRV outlet is no longer used for valve control.
9. Bundle the wiring with zip ties and route it away from the fill line and the antenna. Verify with a multimeter that:
   1. 12 V is present at the relay module coil when SW1 and SW2 are closed;
   2. 12 V is **absent** at the coil when SW2 is open;
   3. the IGN outlet is dead with the relay unpowered;
   4. no continuity exists between the 12 V rail and the 5 V logic rail.
10. Do **not** connect an e-match or an igniter. Proceed to OP 9.

> **Caution.** The igniter outlet is now capable of firing whenever SW2 is closed and the pad node is powered. Treat the IGN outlet as live from the moment SW1 is switched on. Perform all subsequent checkouts with a 12 V lamp or a 100 Ω 5 W resistor in place of the igniter.

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■ NEW OP 9: Assemble and Wire the Control System

QTY	Part Number	Description
1	ESP32-S3-DEVKITC-1	ESP32-S3 devkit — **TX** node (or NUCLEO-H753ZI, see note)
1	DX-LR41-24T12S	2.4 GHz LoRa module
1	ANT-24G-SMA-3DBI	2.4 GHz antenna
1	PRINTED-CARRIER	TX-node carrier (small; may be a 3-pin harness)
1	LAPTOP	Operator laptop with a free USB-C port
1	USB-C-PANEL-1M	Panel extension from OP 6
1	PWRBANK-5V-2A	Optional USB power bank

1. Install PlatformIO on the laptop, then flash the two nodes from the change package:

   ```
   cd firmware
   pio run -e tx-node   -t upload    # first board (ESP32-S3 TX)
   pio run -e pad-node  -t upload   # second board (ESP32-S3 PAD)
   pio run -e tx-nucleo -t upload  # first board, if a NUCLEO-H753ZI is used
   ```

   Flash one board at a time with the other disconnected. Label each board **TX** or **PAD** immediately; the firmware is not interchangeable.

2. Wire the TX node's DX-LR41 exactly as in OP 6 step 2 (3V3, GND, D16→TXD, D17←RXD, D18→AUX), fit the antenna, and connect the node to the laptop through the panel-mounted USB-C extension.

3. Copy the `software/` directory to the laptop. Check that the application starts:

   ```
   python software/run_gui.py --sim
   ```

   The simulated pad must appear in the window and the banner must read `LINK OK — NOT ARMED`. This proves the laptop side before any radio is involved.

4. Close the simulation and start the application for real. The TX node enumerates as a USB CDC device. If the port does not appear, install the ESP32 USB driver for the operating system; the node does not use a CH340 or CP210x bridge.

5. Record the two nodes' MAC addresses and module serials in the GSE log book. They are the only pair in the fleet that will talk to each other; a spare module must be re-provisioned (§Appendix A of the change package).

6. Mount the TX node in the laptop bag or on the lid of the toolbox. Do not mount it inside the toolbox: the antenna must have line of sight to the pad.

Alternative TX node: NUCLEO-H753ZI. The laptop-side node carries no safety state, so it may be built from an ST NUCLEO-H753ZI (MB1364E) instead of the ESP32-S3 without any other change. The node then presents itself as an ST-LINK virtual COM port, which needs no driver. Wire the module to **USART2** — PD5 (CN11 pin 41) to module RXD, PD6 (CN11 pin 43) to module TXD, PD7 (CN11 pin 45) to AUX, 3V3 from CN11 pin 5 and GND from CN11 pin 8 — flash with `pio run -e tx-nucleo -t upload`, and give the application the new port name. The laptop, the pad node, the protocol, the channel map and every procedure in this section are unchanged: both builds run the same bridge code and the same 20 Hz pacing, 200 ms stale-frame rejection and 0.3 s dead-man. Two cautions specific to this board: the VCP is **USART3** (PD8/PD9), not USART2, and the Arduino headers D0/D1 are PG9/PG14 and are **not** a UART.

Harness colour convention

Signal	Pad-node pin	Connector	Note
FILL PWM	D4	3-pin header, signal on the outside pin	matches the vehicle harness order
VEHICLE PWM	D5	3-pin header, feeds the Y-harness	both valves move together
PURGE PWM	D6	3-pin header	optional valve
IGN	D7 → relay	NEMA outlet, longer slot = +12 V	identical to rev 01 outlet wiring
HEARTBEAT	D8	on-carrier only	25 Hz, drives Q1
UART	D16 / D17 / D18	4-pin header	9600 8N1

Why the heartbeat exists. The relay coil is wired in series with a MOSFET that only conducts while the pad node is toggling D8. A firmware hang, a crash loop or a failed flash therefore drops the igniter without relying on any software watchdog. The microcontroller's own task watchdog is a second, slower layer.

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■ NEW OP 10: Node Bring-up, Configuration and Checkout

This operation replaces rev 01 OP 8 (Transmitter Setup and Firmware Configuration) in full. It must be completed, and its results recorded, before any pressure source is connected to the GSE and before any igniter is attached.

Power-up order

1. Power the TX node from the laptop and start the application.
2. Select the TX node's port and press **Connect**. The event log shows `connected`, then `HELLO_OK` once the pad answers.
3. Turn SW1 on. Verify that the banner changes to `LINK OK – NOT ARMED` within one second, that `last valid CMD` is below 100 ms, and that `rejected frames` stays at zero.

> **Caution.** The pad node always boots into the safe state: FILL closed, PURGE closed, VEHICLE open, igniter off. If the read-back does not show that state after power-up, stop and troubleshoot before continuing.

1. Failsafe verification (mandatory, every session)

1. With SW2 open and no igniter connected, close and re-open SW1. The banner must turn red (`PAD SILENT`) within 1.5 s and the read-back must show the safe state.
2. Unplug the TX node from the laptop. The pad must report `COMMS LOSS – PAD SAFE` within 1.5 s.
3. Close the GSE application. The TX node must begin transmitting its own safe commands within 0.3 s, and the pad must remain safe until the application is restarted.
4. Restart the application and confirm `LINK OK – NOT ARMED`.

2. Channel mapping verification

1. Operate each switch in the application and watch the corresponding valve:
   1. FILL → the fill valve opens and closes with Switch position shown as OPEN / CLOSED;
   2. VEHICLE → both vehicle valves move together through the Y-harness;
   3. PURGE → the purge valve moves, only if "purge valve fitted" is ticked.
2. Verify that the read-back panel reports the state the pad actually observed, not the state that was requested.

3. Igniter circuit verification (lamp or resistor only)

1. Fit a 12 V lamp across the IGN outlet.
2. With ARM off, press-and-hold the fire control: the lamp must stay dark and the log must record that the press was inhibited.
3. Close SW2, press **ARM**: the banner turns green.
4. Press-and-hold fire: the lamp lights, the read-back shows `IGNITER ENERGISED`, and the lamp goes out at the configured limit even if the control is still held.
5. Release and press again: the lamp must not re-light until the control has been released.
6. Press `Esc`: ARM drops and the lamp goes dark.

7. Open SW2 with ARM on and fire again: **the lamp must not light**, even though the application reports the igniter as energised. This is the hardware interlock working as designed; note the mismatch in the log book.

4. Servo endpoint adjustment

Perform this step before finally screwing the servo horn to the valve handle, exactly as rev 01 required.

1. Connect only the FILL servo. Ensure no pressure source is connected.
2. Operate FILL. With the valve CLOSED the horn must be perpendicular to the servo body; with the valve OPEN it must be parallel.
3. If either position cannot be reached, adjust **FILL travel** in the application (5–90°) and press **Apply to pad**. The pad echoes `CFG_CHANGED` when it has accepted the value.
4. If the horn still cannot reach the stop at the mechanical limit, remove the horn and re-fit it one spline over. Do not exceed 90° of travel: the ball valve has no over-travel stop.
5. Repeat for VEHICLE (both valves) and PURGE.

5. Configuration to record for the day

Parameter	Default	Range	Recorded value
FILL travel	45°	5–90°	
VEHICLE travel	45°	5–90°	
PURGE travel	45°	5–90°	
Link watchdog	1.0 s	0.3–3.0 s	
Igniter limit	0.5 s	0.1–2.0 s	
Purge enabled	yes/no	—	
Air-rate preset (LEVEL)	3	0–8	
Pad module address / channel	0002 / 0C	—	
TX module address / channel	0001 / 0E	—	

Re-programming is expected. The whole point of this subsystem is that the channel map, endpoints, watchdog and igniter limit are software parameters. Change them from the application, record them above, and re-run steps 1–3 after any change to the failsafe timing or the igniter limit.

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Failsafe behaviour — reference

The pad node re-derives every output, every 20 ms, from two inputs only: the last command frame it accepted, and its own clock. It does not trust the laptop, the radio, or the previous state of its own outputs.

Condition	FILL / PURGE	VEHICLE	IGNITER	Notes
Boot, no frame received yet	CLOSED	OPEN	OFF	Outputs are written before the radio is configured
Normal operation	follows command	follows command	armed + commanded	State read-back at 10 Hz
No valid frame > watchdog (1.0 s)	CLOSED	OPEN	OFF	Primary failsafe; same safe set as the FS-iA10B was configured for
Laptop silent > 0.3 s	CLOSED	OPEN	OFF	TX node synthesises safe commands at 20 Hz
Arm control released	unchanged	unchanged	OFF within 20 ms	Arm code is required in every frame
Igniter limit reached	unchanged	unchanged	OFF, latched	Requires a release and a fresh press
SW2 (arm key) opened	unchanged	unchanged	OFF, immediate	Hardware; independent of firmware
Heartbeat stopped (firmware wedged)	held	held	OFF, immediate	Hardware; the MCU reboots into the boot state within 2 s
Pad power lost	held	held	OFF	Servos hold position without a signal

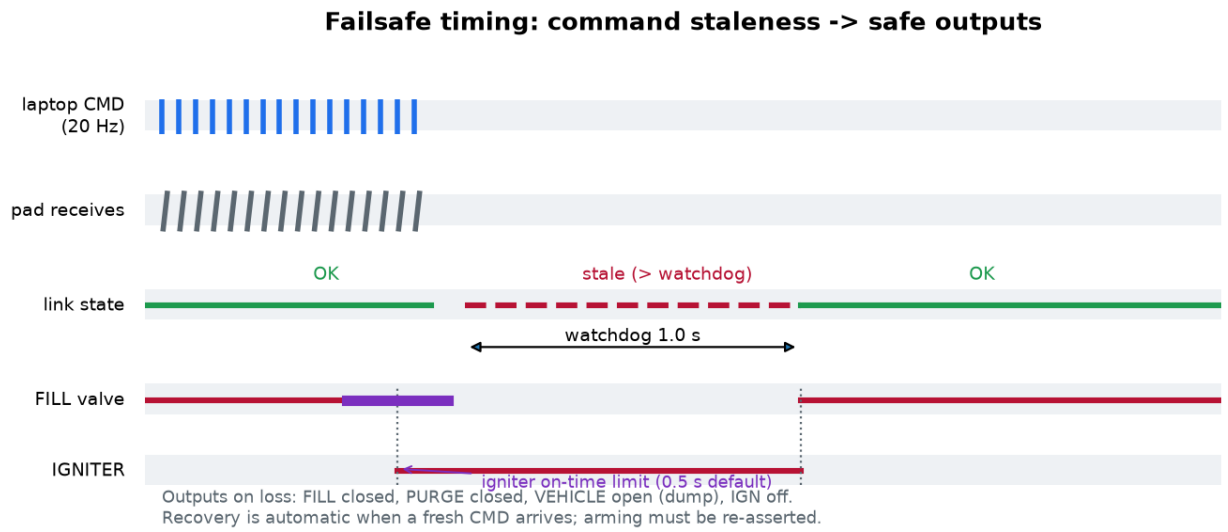


Figure 6.2 (new): Failsafe timing across a one-second link blackout. The command stream stops, the watchdog expires, the outputs snap to the safe set, and normal operation resumes automatically when frames arrive again — but arming must be re-asserted deliberately.

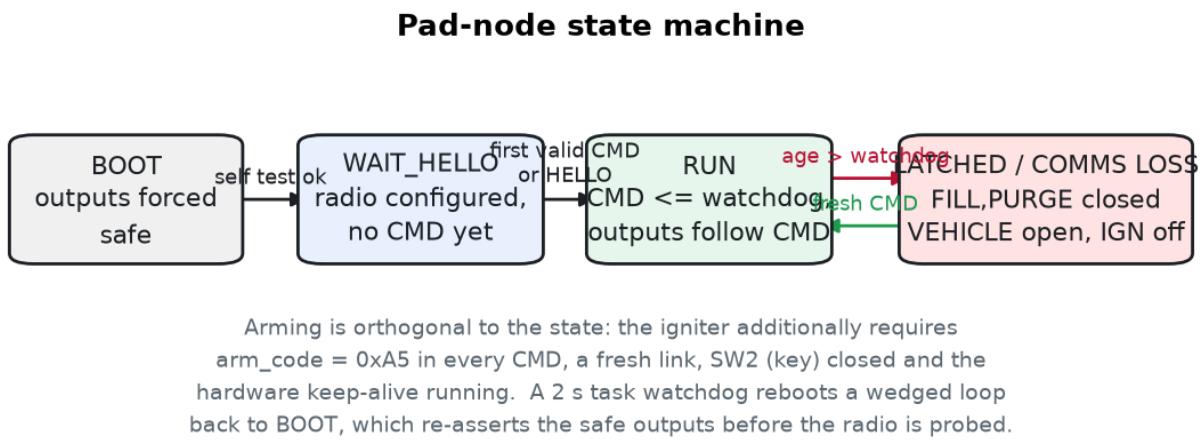


Figure 6.3 (new): Pad-node state machine. Arming is orthogonal to the state: even in RUN, the igniter additionally requires the arm code, a fresh link, SW2 closed and the heartbeat running.

Operator console

The application is deliberately plain: a status banner, the switch positions, the pad's read-back, the configuration panel, and an event log. Everything shown in grey or italic is requested; everything shown solid is reported by the pad. The two must agree before any pressure is admitted to the vehicle.

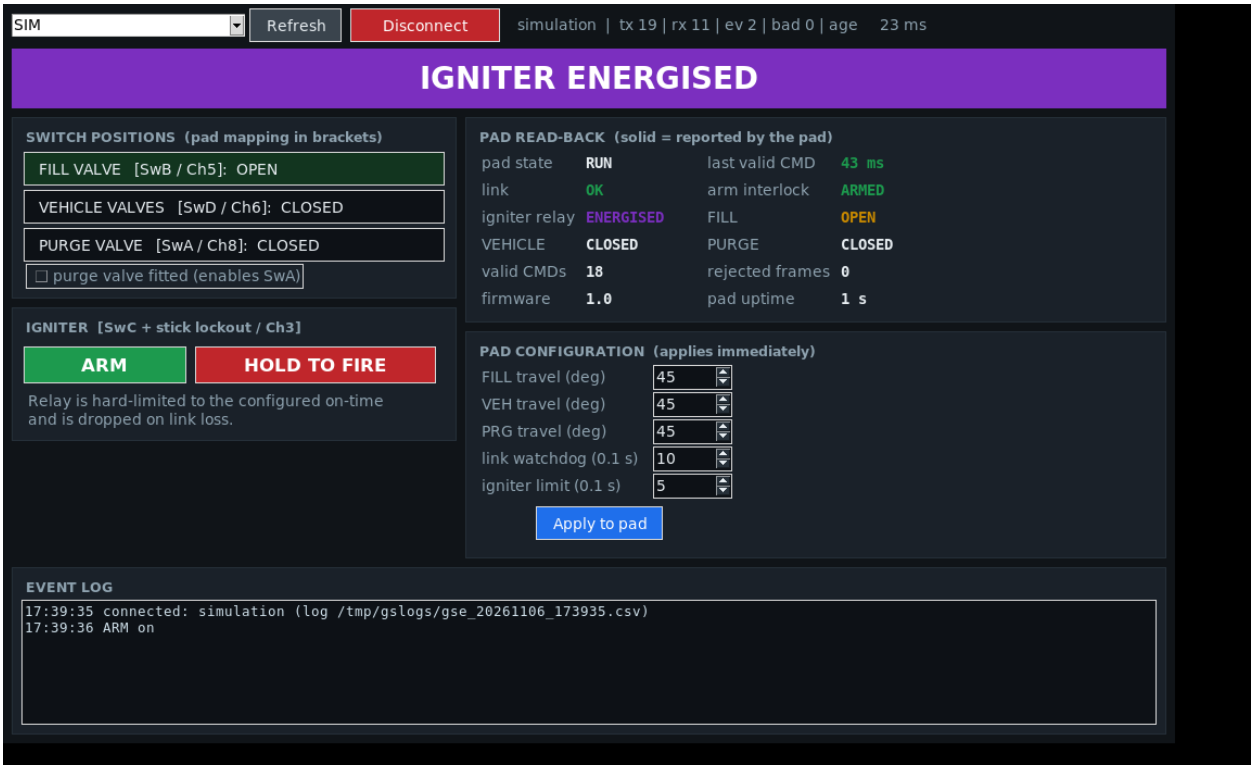


Figure 6.4 (new): GSE control application. The banner is purple because the igniter is energised; the read-back confirms FILL open and VEHICLE closed. In this capture the pad is simulated — the window title says so, and the same window is used for operator training.

Changes to Section VII — Launch Preparation and Procedures

Apply these four edits when the Section VI insert is accepted. They are written as drop-in replacements for the numbered steps they supersede.

§7.2.1 Required items

Item	Rev 01	Rev 02
e	~~Transmitter for GSE control~~	**Operator laptop with the GSE application installed and charged, the TX-node carrier, and one spare USB-C cable.** Keep the laptop in a padded case; it is now a flight-critical item.

§7.2.2 Ground system setup

Replace step 18 with:

1. Optional: start the GSE application on the laptop, connect to the TX node, and confirm the banner reads `LINK OK — NOT ARMED` and that `rejected frames` is zero. Do not arm, and leave SW2 open.

§7.3.2 GSE startup and checkouts

Replace steps 1–3 ("Power on Transmitter & GSE") with:

1. Start the GSE application and connect to the TX node.
2. Turn SW1 on. Verify that the transmitter has established a connection with the pad node within a few seconds, that the banner reads `LINK OK — NOT ARMED`, and that `last valid CMD` is below 100 ms.
3. Confirm SW2 (arm key) is open and the key is in the operator's pocket.

Replace steps 7–11 with the same functional checks, operated from the application:

1. Command FILL open. Visually confirm that the valve has opened; the read-back must show OPEN.
2. Command FILL closed. Visually confirm; the read-back must show CLOSED.
3. Command VEHICLE open, then closed, confirming both valves follow.
4. Spark check, with SW2 open and the igniter leads touching each other: no spark. Close SW2, press ARM, press-and-hold the fire control: a spark within 0.1 s. Release, then press `Esc`: no spark, and ARM is off in the read-back.

§7.3.3 Vehicle arming

Replace step 3 in part and add one step:

1. Confirm that both deployment controllers report continuity of both drogue and main ejection charges. Confirm that the Operator has physical possession of the laptop and the arm key, that ARM is off, that all switch positions are reported CLOSED or SAFE, and that the link banner is green or amber — never red.

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Add after step 7 (oxidizer supply valve open):

1. Close SW2 and keep the key with the Operator. SW2 must be opened again before any person approaches the pad, and before the vehicle valves are commanded open for a dump.

§7.5.2 Aborted firing — step 4a

Replace the caution about not cycling the transmitter with:

a. During this time, do not close and re-open the GSE application. Restarting the application re-runs its link handshake; although the pad stays safe while the link is down, an accidental re-arm must be avoided. Do not open SW2 either: the vehicle valves must remain commandable. If three attempts to remotely open the vehicle valves are unsuccessful, close the application and leave it closed until the vehicle and pad are completely safed.

§7.5.3 Vehicle shutdown

Add to step 1:

Turn off both deployment controllers, then press `Esc` on the application to drop ARM and release the igniter, then open SW2 and remove the key. Insulate the altimeter power leads to prevent accidental re-arming.

Appendix — Troubleshooting the control link

Symptom	Likely cause	Action
Application cannot find the port	TX node in bootloader mode, or a charge-only USB cable	Hold BOOT, tap RESET, re-flash; try a known data cable
Banner stays `NO PAD TELEMETRY`; pad shows an age of 65535	Pad not powered, modules not on matching LEVEL/channel, or KEY mismatch	Check SW1; open a serial terminal on the pad node at 115200 and read the boot message; re-provision both modules with `RADIO CONFIG`
`rejected frames` climbing	RF interference, range, or a corrupted module setting	Move the antennas clear of metal; reduce LEVEL to 5 or 7 on both ends; check `AT+HELP` banners for identical LEVEL/KEY
Frequent `COMMS_LOSS` at 100 ft but not at 5 ft	Path obstruction or antenna polarisation	Re-orient both antennas vertically; raise the TX node; repeat the range test before proceeding
Pad reports RUN but ignores switch changes	Laptop clock stalled or the application lost the port	Re-connect; confirm the CSV log is still being written
Servo buzzes at the end of travel	Travel set beyond the mechanical stop	Reduce travel for that channel by 5° and re-apply; re-fit the horn if needed
Igniter will not fire, read-back says energised	SW2 open (hardware interlock), or the igniter limit latched after a previous press	Close SW2; release and press the fire control again
Igniter fires on power-up	Firmware fault — **stop**	Do not fly. Re-flash, repeat the bench ignition test, and report the log file
PWM relay or servo twitches when the relay operates	Coil transient coupling	Verify the flyback diode orientation; add 100 nF across the relay contacts; separate the coil wiring from the servo harness

Spares to carry

- One spare DX-LR41-24T12S with a pre-provisioned copy of the pad configuration
- One spare ESP32-S3, flashed as PAD
- One spare 2.4 GHz antenna
- One spare USB-C data cable, and the USB power bank
- The FS-i6X and FS-iA10B, boxed, with the rev 01 OP 8 card — the rollback path if the new link repeatedly fails its checkouts

Note on spares. A spare pad node must have the same firmware version as the installed one, and its module must be provisioned to the same address and channel. Label spares with the firmware version and the provisioning date.

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